

KOA MARIE GERVAIS NELLY

Lead AI Engineer & Data Architect | Founder @ Archi Cam AI

EXECUTIVE SUMMARY

Lead AI Engineer & Data Architect (Google Developer Program Member), leveraging Google Antigravity to design neuro-symbolic autonomous multi-agent architectures, Neo4j GraphRAG, and sovereign MLOps pipelines. Expert in AI ethics (transparency, SHAP explainability, EU AI Act compliance, deterministic zero-hallucination models). Founder of Archi Cam AI (official Google Africa Applied AI Lab applicant 2026), bridging civil engineering methodology, mathematical rigor, and bold product vision.

FLAGSHIP AI PROJECTS

Archi Cam AI — *Agentic AI SaaS & 5D BIM*

- Official Applicant Google Africa Applied AI Lab. Built with Google Antigravity (Gemma 4 12B, Gemini 2.5, BAEL 91).
- Automated Excel BOQs in <45s (−99.2% time, MLflow $R^2=0.9872$) and 3D renders via Imagen 3 + ControlNet.
- IfcOpenShell integration for 5D BIM models: costs + schedules automated via LangGraph agents.

K1-MATHINFO (v3.0.0) — *Sovereign Multi-Agent AI, WikiSkills & OKF Certification*

- Sovereign DMI system (Univ. of Ngaoundéré): 470 theses, 19 M1 projects, 1,366 Neo4j nodes.
- 6 LangGraph agents orchestrated via WikiSkills (critic anti-hallucination, Cypher optimizer), RRF k=60, SHA-256 No-LLM (100%).

Sovereign.BI Agentic — *Enterprise Security & Agentic BI*

- NL-to-SQL/Graph engine (PostgreSQL pgvector, Neo4j N10S, <5s latency) with ABAC guardrails + SHAP Sentinel.
- Executive dashboards via Streamlit & FastAPI — deployed on sovereign cloud Docker infrastructure.

Dataset Automator & VigieSahel — *MLOps Pipeline & Climate AI Impact*

- Dataset Automator: Google Antigravity (Google Cloud Hackathon) — TabFM, BigQuery DataFrames, EU AI Act.
- VigieSahel: −35% crop loss, +14d epidemic forecast (XGBoost $R^2>94\%$, Supabase, MLflow).

PROFESSIONAL EXPERIENCE

AI Lead & Data Science Consultant — *2025 – Present*

Independent Projects & Enterprises | Douala, CM

- Ethical sovereign AI systems: GraphRAG Neo4j pipelines, high-dimensional EDA, multi-source RAG architectures.
- Executive KPI dashboards, automated reporting, and SHAP explainability audit for Cameroonian SMEs.

Aviation Security Officer (AVSEC) — *2018 – Present*

CCAA — Cameroon Civil Aviation Authority | Douala, CM

- Threat assessment, secure access control, and regulatory compliance audits (ICAO Annex 17).
- Operational crisis management: team coordination in emergency situations, anti-intrusion protocols.

EDUCATION & CERTIFICATIONS

M.Sc. in Applied Artificial Intelligence — *2025 – 2027 [In Progress]*

University of Ngaoundéré | Cameroon

- ML & Bayesian Statistics, Data Engineering & Neo4j, Computer Vision & Robotics, Ethics & Cybersecurity, Production MLOps.
- Research project: K1-MATHINFO v3 — multi-source agent certified OKF v0.2 (SHA-256 No-LLM).

B.Sc. in Civil Engineering (Building Option) — *2015 – 2016*

ISTDI / IUC Douala | Cameroon

- Structural calculations (BAEL 91), quantity surveying, construction project management — AI-applied estimation base.

HONORS & APPLIED AI DISTINCTIONS

- Google Africa Applied AI Lab — Accra 2026** — Official candidacy — Archi Cam AI platform (5D BIM + GenAI).
- Google Cloud #AllThingsAgentic Hackathon** — Dataset Automator v4.0 — Google Antigravity, TabFM, bigframes, WIT, MCT.
- Google Developer Program · Devpost** — Active member — open-source AI contributions, hackathons & ambassador program.



LEAD AI ENGINEER

Founder · Archi Cam AI

CONTACT & PROFILES

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- linkedin.com/in/marie-gervais-koa
- devpost.com/magenel85
- Google Developer Program

AI & LLM STACK

- Google Antigravity IDE ■■■■■■
- LangGraph & WikiSkills ■■■■■■
- Google Gemma 4 (12B) ■■■■■■
- Gemini 2.5 / 1.5 Pro ■■■■■■
- Google TabFM (Tabular) ■■■■■■
- Firebase Genkit ■■■■■■
- Neo4j GraphRAG — Agent K1

DATA & GRAPHS

- Neo4j / Cypher Graph ■■■■■■
- PostgreSQL / pgvector ■■■■■■
- Google BigQuery ■■■■■■
- Supabase Realtime ■■■■■■
- Pandas / NumPy ETL ■■■■■■

DEV & MLOPS

- Python 3.11+ / MLOps ■■■■■■
- FastAPI / Next.js 14 ■■■■■■
- MLflow & Data Drift ■■■■■■
- SHAP Sentinel Audit ■■■■■■
- Docker & Streamlit ■■■■■■
- IfcOpenShell (5D BIM) ■■■■■■

ETHICS, SECURITY & AUDIT

- AI Ethics & Anti-Hallucination
- Quorum 4-Eyes Governance
- OKF v0.2 SHA-256 No-LLM
- EU AI Act Compliance (RSASSA)
- AVSEC (ICAO Annex 17)

LANGUAGES

- French — Native / Fluent
- English — Technical / Pro

KEY ASSETS

- Dual Competence: AI & Civil Eng.
- Aviation Security & Crisis (CCAA)
- Mathematical Rigor & Guardrails